# Md. Abdullah Al Mamun

Graduate Researcher, ECE, Temple University • Quantum Algorithms, Quantum ML & Quantum Error Correction

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## RESEARCH PROFILE

Quantum computing researcher focused on **when quantum models actually help and what limits them**: trainability and noise diagnostics for variational circuits on **IBM Quantum hardware**, GPU-scale studies of **trainability versus classical simulability** with CUDA-Q, and **noise attribution for early-fault-tolerant neutral-atom protocols**. Author of **HilbertBench**; contributor to QuEra's open-source **tsim** sampler; lead author in deep learning for computational pathology; former industry AI engineer. AI-native, with results kept verifiable through pre-registration and tests.

## EDUCATION

**Temple University, Philadelphia, PA** Jan 2025 – Present

M.S. in Electrical Engineering *(expected Dec 2026; one course remaining)* — **GPA 3.78 / 4.00**

Ph.D. program in Electrical & Computer Engineering (M.S. en route)

**University of Rajshahi, Bangladesh — B.Sc. in Computer Science & Engineering** 2018 – 2023

Thesis: Seizure detection from EEG (ResNet; 50+ controlled experiments)

## PUBLICATIONS & MANUSCRIPTS

1. **M. A. Al Mamun**, et al., "HilbertBench: A Backend-Agnostic Framework for Quantum Machine Learning Diagnostics with Validation on Quantum Hardware," manuscript in preparation for *Quantum*, 2026. IN PREP
2. S. Purba, **M. A. Al Mamun**, I. Obeid, & J. Picone, "No Quantum Advantage Without Structure: A Controlled Benchmark of Quantum Machine Learning on Classical Data," *IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, Dec. 2026. ACCEPTED
3. **M. A. Al Mamun**, S. Purba, I. Obeid, & J. Picone, "Parameter-Efficient Fine-Tuning of Pathology Foundation Models for Microsegmentation of Breast Whole-Slide Images in Digital Pathology," *IEEE SPMB*, Dec. 2026. ACCEPTED
4. **M. A. Al Mamun**, R. C. Purba, I. Obeid, & J. Picone, "Automated Interpretation and Visualization in Digital Pathology," *Proc. 52nd Northeast Bioengineering Conference (NEBEC)*, 2026.
5. B. Hackel, A. Bagritsevich, A. Dumitrescu, **M. A. Al Mamun**, R. C. Purba, D. Heathcote, I. Obeid, & J. Picone, "Enabling Microsegmentation: Digital Pathology Corpora for Advanced Model Development," in *Signal Processing in Medicine and Biology*, Springer, 2026.
6. C. Dumitrescu, S. Purba, **M. A. Al Mamun**, et al., "Rapid and Inexpensive Precision Breast Cancer Screening Using Machine Learning," 2025.

## QUANTUM RESEARCH & OPEN-SOURCE SOFTWARE

**Star on Atoms — which noise channel limits transversal STAR on neutral atoms** Oct 2026 – Present

*github.com/mamuncseru/star-on-atoms • pre-registered protocols • tsim, Bloqade, Stim*

- Attributed the injection error of **QuEra's transversal STAR architecture** (post-selected small-angle rotations for early fault tolerance) to individual physical noise channels, using QuEra's published circuits and hardware model.
- Committed an exact first-order fault enumeration as a **pre-registered prediction**; confirmed it blind on **802 million sampled shots** at code distances 3, 5 and 7, with 95% Wilson intervals.
- Single-qubit gates carry 43–52% of the injection error; halving their error buys about $1.3\times$ in simulation volume at $d = 5$.
- Found a decoding pitfall and contributed the fix upstream: **QuEraComputing/tsim pull request #234**, merged Nov 2026.

**HilbertBench — non-intrusive diagnostics for quantum ML experiments** 2025 – Present

*github.com/mamuncseru/hilbertbench • hilbertbench.readthedocs.io • DOI 10.5281/zenodo.21009244 •* `pip install hilbertbench`

- Designed and built a framework that wraps a single Qiskit V2 **Estimator/Sampler** or PennyLane device and records every circuit execution into an append-only, SHA-256-sealed trace, with no changes to algorithm code and no extra shots.
- Implemented six analyzers computed from the trace: **barren-plateau** detection (gradient variance with CIs), shot-noise ratio, optimizer convergence, circuit structure, **KL expressibility**, and hardware noise profile.
- Ran diagnostics on **IBM Quantum processors** via `EstimatorV2(mode=backend)`; 319 documented tests, CI, four end-to-end tutorials on real QML failure modes; basis of a journal manuscript for *Quantum*.
- Built a **CUDA-Q integration** and ran a GPU pilot on an **NVIDIA A40**: 19,200 recorded executions (4–26 qubits, fp64). Linear-depth local-cost circuits lost gradient variance only about $4\times$ from 4 to 26 qubits while the MPS bond dimension needed to reproduce them rose from 8 to 64 (8 to 20 qubits); recording overhead under 0.5%.

**Controlled Benchmark of QML vs. Classical Learning (co-author, IEEE SPMB 2026)** 2026

- Benchmarked quantum machine learning models against strong classical baselines on classical data under matched, controlled conditions, showing that performance gains require problem structure the quantum model can exploit.

**Quantum Advantage Quest — open research program on provable quantum advantage** 2026 – Present

*github.com/mamuncseru/quantum-advantage-quest • project site*

- Problem-first search for quantum advantage with pre-registered kill criteria; every claim is attacked with the strongest classical tools first (e.g., Tang-style dequantization, BBBV lower bounds).
- 13 algorithm "autopsies" (Deutsch–Jozsa → Decoded Quantum Interferometry, quantum Gibbs sampling) and a Problem-Shape Catalog mapping problem structures to quantum mechanisms and to what kills them.
- Tested statevector simulator (NumPy/SciPy): Shor factoring 35 on 19 qubits, glued-trees quantum walk separation, Davies-generator Gibbs sampler, QSVT Chebyshev identity.

## RESEARCH EXPERIENCE

**Graduate Research Assistant — Temple University** Jun – Aug 2025; Jan 2026 – Present

- Quantum ML for the NSF FET project *Quantum Computing Based Approach to Undirected Generative ML Models*: QML benchmarking, trainability diagnostics, and hardware experiments (Qiskit, PennyLane).
- Lead author on parameter-efficient fine-tuning of pathology foundation models for microsegmentation of gigapixel breast whole-slide images (IEEE SPMB 2026); applying reinforcement learning to digital pathology.
- Built GPU-accelerated WSI pipelines (EfficientNet, TransMIL, self-supervised learning) on SLURM HPC clusters.

**Research Intern — Kyushu Institute of Technology, Japan** Feb 2023 – Mar 2023

Sakura Science Exchange Program. EEG acquisition under controlled protocols; preprocessing and analysis in MATLAB.

## INDUSTRY EXPERIENCE

**AI Engineer — My MedicalHub International Ltd, Dhaka, Bangladesh** Aug 2023 – Dec 2024

- Led real-time computer-vision models for medical diagnostics (classification, detection, segmentation, pose estimation) in PyTorch/TensorFlow; deployed and scaled on AWS and Azure with monitoring and retraining workflows.

**Research Assistant (part-time, remote) — My MedicalHub International Ltd, Tampa, FL** Aug 2020 – Jul 2023

Built and annotated medical datasets; object detection, segmentation, and pose estimation with systematic evaluation.

## TEACHING & COMMUNITY

**Instructor — IEEE Quantum Machine Learning Workshop (QMLW 2026)** Summer 2026

Designed and delivered hands-on QML lectures and coding exercises (Qiskit, PennyLane) for signal-processing and biomedical applications.

**Graduate Teaching Assistant — Temple University** Jan 2025 – Dec 2025

Engineering Computation I & II (ECE 1111, ECE 3822); Electrical Devices & Systems I (ECE 2112).

**Attendee — IEEE Quantum Week 2026 (QCE26), Toronto, Canada** Sep 2026

**Participant — Microsoft Quantum QDK Workshop, New York, NY** May 2026

## TECHNICAL SKILLS & HONORS

**Quantum:** Qiskit (V2 primitives, IBM Quantum hardware), PennyLane, CUDA-Q (`nvidia`, `tensornet-mps`), Stim, tsim, Bloqade, Microsoft QDK; variational circuits, barren plateaus, expressibility, shot-noise and hardware-noise analysis, QEC simulation and detector error models, quantum walks, Gibbs sampling, QSVT, dequantization.

**Machine learning:** PyTorch, PyTorch Lightning, TensorFlow; foundation models & parameter-efficient fine-tuning, multiple-instance learning, self-supervised learning, computer vision, reinforcement learning.

**Scientific computing & tools:** Python, NumPy, SciPy, SymPy, pandas, C++, MATLAB; pytest, Git/GitHub Actions, Docker, Linux, SLURM/HPC, CUDA, MkDocs/Read the Docs, Quarto, LaTeX, AWS, Azure.

**AI-native workflow:** coding agents (e.g., Claude Code), checked by pre-registered predictions, tests, and logged agent use.

**Honors:** Sakura Science Exchange Program, JST (2023) • Microdegree in AI, Fusemachines (2022–23) • IEEE Member